

ST CLAIR COUNTY INTL, MI, USA

WMO: 725384

Lat: 42.911N

Lon: 82.529W

Elev: 640

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04888

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.5	4.6	-8.8	3.5	-0.1	-3.8	4.6	6.2	20.8	27.3	18.9	26.4	3.5	300	0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	89.6	73.4	85.7	71.4	82.3	69.6	75.7	84.1	73.8	81.7	72.3	79.9	8.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	125.1	79.7	71.9	120.9	78.7	70.0	113.1	77.0	39.6	83.9	37.8	81.3	36.3	79.1	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.5	16.5	14.5	DB	-9.7	93.3	8.2	3.5	-15.6	95.8	-20.4	97.9	-25.0	99.9	-30.9	102.4	
			WB	-9.0	79.3	8.8	2.4	-15.4	81.0	-20.5	82.4	-25.5	83.7	-31.9	85.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	24.4	25.4	34.0	45.2	56.5	65.3	69.6	68.0	61.6	50.6	39.5	30.1
	DBStd	18.21	11.19	11.03	10.65	8.81	8.43	7.00	5.90	5.48	7.33	8.43	8.50	9.03
	HDD50	3269	793	688	511	196	29	1	0	0	4	98	331	618
	HDD65	6799	1258	1107	963	596	289	81	22	32	150	455	765	1081
	CDD50	2408	0	1	13	52	231	460	608	557	353	115	16	2
	CDD65	462	0	0	0	2	26	90	165	124	48	7	0	0
	CDH74	4617	0	0	6	39	336	973	1621	1094	483	65	0	0
	CDH80	1310	0	0	0	6	81	306	506	268	130	13	0	0
Wind	WSAvg	6.0	7.1	6.9	7.1	7.5	6.1	4.9	4.2	4.1	4.6	5.8	6.7	6.6
Precipitation	PrecAvg	33.7	2.3	1.9	2.4	3.0	3.7	3.0	3.4	3.1	3.3	2.6	2.6	2.4
	PrecMax	47.0	3.9	3.9	5.1	6.2	7.6	5.9	6.7	6.1	7.0	6.7	5.5	4.4
	PrecMin	24.7	0.7	0.5	0.9	0.8	1.6	0.9	0.6	0.6	0.6	0.6	0.8	0.8
	PrecStd	4.8	0.9	0.9	1.1	1.3	1.5	1.4	1.6	1.2	1.9	1.3	1.1	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	57.3	72.5	79.4	86.3	90.7	91.5	90.5	89.9	81.2	66.2	57.3
		MCWB	51.7	50.2	59.3	61.9	68.9	74.0	74.6	74.7	72.3	68.0	56.6	53.5
	2%	DB	48.4	49.8	63.2	72.4	81.8	87.8	89.6	85.8	83.7	74.6	61.3	52.1
		MCWB	46.2	45.7	53.0	57.6	66.7	72.0	74.3	72.4	70.4	63.7	54.7	49.3
	5%	DB	43.3	44.7	56.5	66.3	77.4	83.7	85.5	82.2	79.9	70.1	57.1	46.1
		MCWB	39.8	40.1	49.0	54.5	64.0	70.0	71.9	70.2	67.5	60.8	51.4	42.9
	10%	DB	38.7	39.0	50.2	61.3	72.6	80.6	82.0	80.8	75.2	64.5	53.9	42.7
		MCWB	36.1	35.4	44.5	51.4	61.0	68.0	70.0	69.1	65.1	58.0	48.9	39.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.7	53.4	60.5	65.6	72.7	76.3	79.6	77.5	76.0	69.8	59.7	55.0
		MCDB	54.0	55.5	70.6	75.4	82.1	87.0	89.3	86.8	84.8	77.0	63.5	57.7
	2%	WB	46.4	45.9	54.6	60.1	69.2	73.8	76.2	74.7	72.7	66.3	56.0	49.6
		MCDB	47.4	48.6	60.4	68.4	78.2	83.2	84.3	81.8	79.8	71.8	60.5	51.7
	5%	WB	40.7	39.7	50.0	56.3	66.5	72.0	74.0	72.8	70.1	62.5	52.6	43.7
		MCDB	43.2	43.2	56.3	64.7	74.5	80.8	81.7	79.7	75.6	67.4	56.6	45.8
	10%	WB	36.7	36.2	44.7	52.5	63.2	69.8	72.3	71.1	67.6	59.3	48.9	39.4
		MCDB	38.8	39.2	49.3	60.3	70.4	77.1	79.9	77.8	73.4	64.6	52.7	42.4
Mean Daily Temperature Range	5% DB	MDBR	14.2	16.0	17.2	20.1	22.2	22.4	22.1	21.2	22.5	19.3	15.2	12.2
		MCDBR	16.3	19.8	26.5	29.4	27.9	26.7	24.9	23.9	26.9	26.1	20.6	16.7
		MCWBR	14.8	16.2	17.8	17.5	15.3	13.4	12.0	12.2	14.3	15.3	15.7	14.5
	5% WB	MCDBR	15.6	18.6	24.5	25.7	24.4	23.1	21.2	20.3	22.1	21.2	18.6	15.7
		MCWBR	14.9	16.0	18.0	17.8	15.1	13.0	11.8	11.6	13.2	14.2	15.5	14.8
Clear-Sky Solar Irradiance	taub		0.308	0.323	0.340	0.386	0.410	0.434	0.422	0.411	0.379	0.358	0.319	0.295
	taud		2.345	2.295	2.369	2.267	2.253	2.219	2.277	2.295	2.372	2.408	2.485	2.444
	Ebn at Noon		260	276	286	279	274	267	269	268	269	261	257	257
	Edh at Noon		26	32	34	41	43	44	41	39	34	29	23	22
All-Sky Solar Radiation	RadAvg		418	691	1096	1451	1704	1934	1908	1658	1329	851	512	353
	RadStd		55	68	109	126	145	110	98	74	116	75	65	33

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page